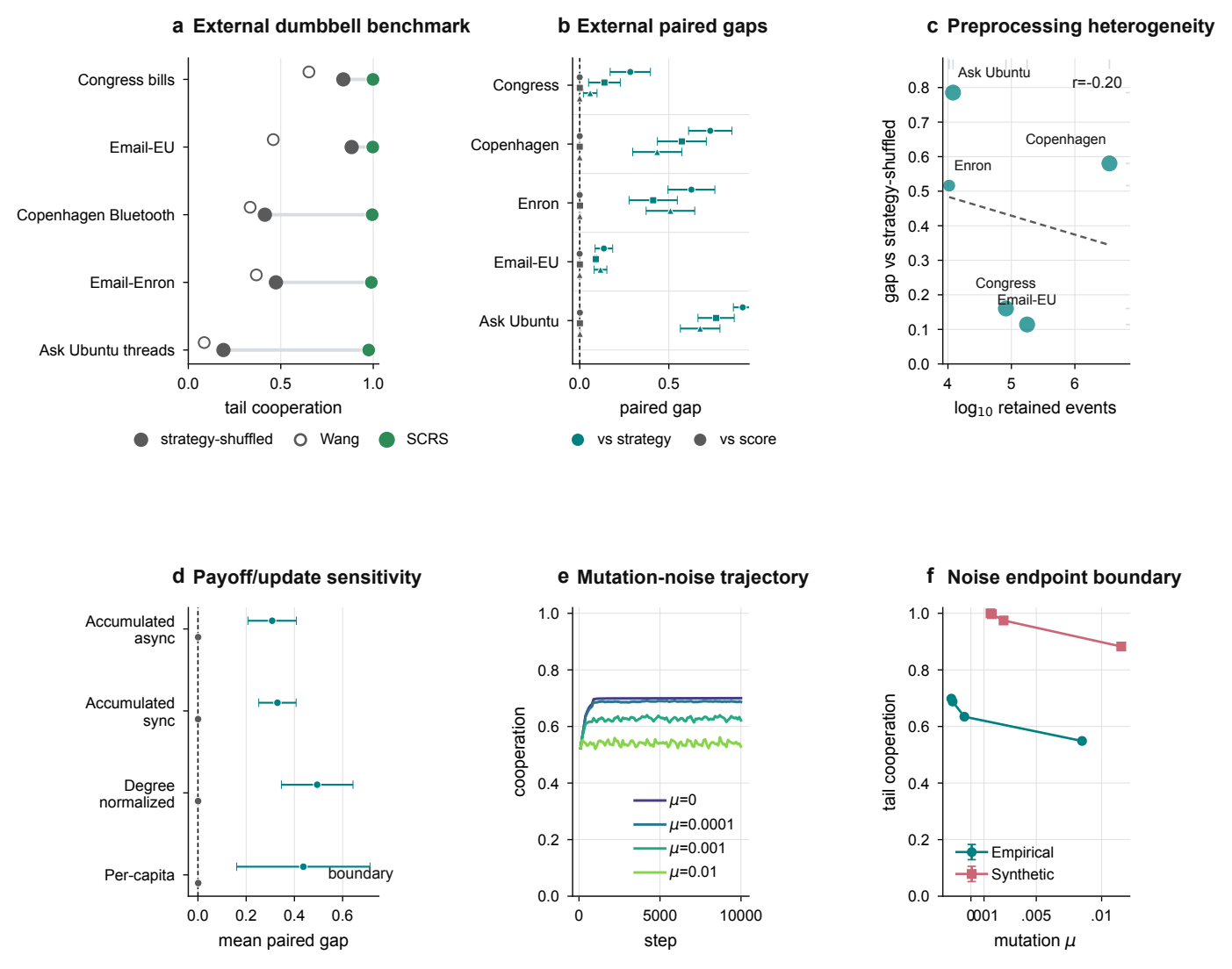

a External dumbbell benchmark
Congress bills
Email-EU
Copenhagen Bluetooth
Email-Enron
Ask Ubuntu threads
0.0
0.5
1.0
tail cooperation
strategy-shuffled
Wang
SCRS
b External paired gaps
Congress
Copenhagen
Enron
Email-EU
Ask Ubuntu
0.0
0.5
paired gap
vs strategy
vs score
c Preprocessing heterogeneity
Ask Ubuntu
r=-0.20
Copenhagen
Enron
Congress
Email-EU
gap vs strategy-shuffled
log10 retained events
d Payoff/update sensitivity
Accumulated async
Accumulated sync
Degree normalized
Per-capita
boundary
mean paired gap
e Mutation-noise trajectory
cooperation
μ=0
μ=0.0001
μ=0.001
μ=0.01
step
f Noise endpoint boundary
tail cooperation
Empirical
Synthetic
mutation μ